Indirect reciprocity with inertial reputation updating

Methods

Methods

Donation game, strategies and private assessment

We consider an infinitely large, well-mixed population in which randomly chosen donors and recipients play a donation game. A donor either cooperates, paying a cost $c > 0$ to give the recipient a benefit $b > c$, or defects, which has no payoff consequences. Individuals use one of three strategies $I \in \{\mathrm{ALLC}, \mathrm{ALLD}, \mathrm{DISC}\}$ with frequencies f_I, $\sum_I f_I = 1$. ALLC intends to cooperate with every recipient, ALLD intends to defect against every recipient, and DISC intends to cooperate if and only if it regards the recipient as good. All individuals share one social norm and one reputation mechanism (the depths k_1, k_2 and readout parameters α, β defined below); strategy evolution changes only the frequencies f_I.

Reputations are private [1, 2, 3]: every individual keeps its own record of every other individual. In each interaction, an observer drawn independently of donor and recipient assesses the donor. A DISC donor chooses its action using its own image of the recipient, whereas the observer judges the donor's realized action using the observer's own image of the recipient, so the two can disagree. We use second-order norms [4]. Cooperation with a recipient regarded as good is assessed positively and defection against such a recipient negatively; towards a recipient regarded as bad, cooperation and defection are assessed positively with probabilities p and q, respectively. Scoring (SC), Shunning (SH), Stern Judging (SJ) and Simple Standing (SS) correspond to $(p,q) = (1,0)$, $(0,0)$, $(0,1)$ and $(1,1)$.

Intended cooperation fails and becomes defection with probability e_1; intended defection is always implemented. After applying the norm, the observer records the opposite assessment with probability e_2. The two errors are independent, and benefits and costs accrue only for realized cooperation. Let $P^+_{A|Y}$ denote the probability that the recorded assessment is positive, given the donor's intended action $A \in \{\mathrm{C}, \mathrm{D}\}$ and the observer's image $Y \in \{\mathrm{G}, \mathrm{B}\}$ of the recipient. With $\varepsilon = (1-e_1)(1-e_2) + e_1 e_2$,

$$\begin{aligned} P^+_{\mathrm{C|G}} &= \varepsilon, \quad & P^+_{\mathrm{C|B}} &= e_2 + (1-2e_2)\big[(1-e_1)p + e_1 q\big], \\ P^+_{\mathrm{D|G}} &= e_2, \quad & P^+_{\mathrm{D|B}} &= e_2 + (1-2e_2)q \end{aligned} \tag{M1}$$

(Supplementary Note 2). Implementation error is already averaged into Eq. (M1). If the observer regards the recipient as good with probability r, intended cooperation and intended defection are therefore assessed positively with probabilities

$$\Phi_{\mathrm{C}}(r) = rP^+_{\mathrm{C|G}} + (1-r)P^+_{\mathrm{C|B}}, \qquad \Phi_{\mathrm{D}}(r) = rP^+_{\mathrm{D|G}} + (1-r)P^+_{\mathrm{D|B}}, \tag{M2}$$

and their difference, the assessment contrast, is $\Delta\Phi(r) = \Phi_{\mathrm{C}}(r) - \Phi_{\mathrm{D}}(r) = \kappa[r + (p-q)(1-r)]$ with $\kappa = (1-e_1)(1-2e_2)$. Unless stated otherwise, $b = 5$, $c = 1$ and $e_1 = e_2 = 0.02$.

Inertial reputation records and binary readout

Observer i holds a latent record L_{ij} of target j in the state space

$$\mathscr{L} = \{\mathrm{G}, \mathrm{W}_1, \ldots, \mathrm{W}_{k_1}, \mathrm{B}, \mathrm{S}_1, \ldots, \mathrm{S}_{k_2}\}, \qquad k_1, k_2 \in \{0, 1, 2, \ldots\}. \tag{M3}$$

After each interaction, only the observer's record of the donor changes, and only through the recorded assessment. A negative assessment moves $\mathrm{G} \to \mathrm{W}_1 \to \cdots \to \mathrm{W}_{k_1} \to \mathrm{B}$ and leaves B unchanged; a positive assessment moves $\mathrm{B} \to \mathrm{S}_1 \to \cdots \to \mathrm{S}_{k_2} \to \mathrm{G}$ and leaves G unchanged. A positive assessment returns every warning state W_m directly to G, and a negative assessment returns every recovery state S_n directly to B (direct reset). If $k_1 = 0$, a negative assessment moves G directly to B; if $k_2 = 0$, a positive assessment moves B directly to G. Losing good standing therefore requires $k_1 + 1$, and regaining it $k_2 + 1$, consecutive assessments of the same target by the same observer. We call k_1 the tolerance depth, k_2 the strictness depth and $K = k_1 + k_2$ the total depth; a record has $K + 2$ states (complete transition table in Supplementary Note 3 and Supplementary Table S5).

Actions and assessments use a binary image that is sampled from the record whenever the record is consulted. The image is good with probability

$$\sigma(\mathrm{G}) = 1, \qquad \sigma(\mathrm{W}_m) = \alpha^m, \qquad \sigma(\mathrm{B}) = 0, \qquad \sigma(\mathrm{S}_n) = 1 - \beta^n, \qquad 0 \le \alpha, \beta \le 1. \tag{M4}$$

Consultations are independent given the current state and do not change it. The tolerance strength α sets how often a warning state is still treated as good, and the strictness strength β how often a recovery state is still treated as bad. We compare three settings: no inertia, $(k_1,k_2)=(0,0)$; full one-step inertia, $(k_1,k_2)=(1,1)$ with $\alpha=\beta=1$; and the selected multi-step architectures with full readout, $(2,0)$, $(3,0)$, $(4,3)$ and $(1,2)$ for SC, SH, SJ and SS (selected from simulated reputation statistics; see below). Setting $\alpha=\beta=0$ at any depths also yields instantaneous updating, because the image then always equals the last recorded assessment. At one-step depths, $(\alpha,\beta)=(1,0)$ and $(0,1)$ give tolerance-only and strictness-only readout.

Stationary reputation map

Consider one record whose successive recorded assessments are independent and positive with probability u. Its state then follows a Markov chain with transition matrix $T(u)=uT^{+}+(1-u)T^{-}$, where T^{+} and T^{-} encode the deterministic updates after positive and negative assessments. For $0<u<1$ the chain is irreducible and aperiodic. Write $v=1-u$. Because W_m can be entered only from W_{m-1} (with $\mathrm{W}_0\equiv\mathrm{G}$) after a negative assessment, and S_n only from S_{n-1} (with $\mathrm{S}_0\equiv\mathrm{B}$) after a positive one, the stationary distribution satisfies $\pi_{\mathrm{W}_m}=v^m\pi_{\mathrm{G}}$ and $\pi_{\mathrm{S}_n}=u^n\pi_{\mathrm{B}}$. The only transitions from the good-standing states $\{\mathrm{G},\mathrm{W}_m\}$ to the bad-standing states $\{\mathrm{B},\mathrm{S}_n\}$ complete a run of k_1+1 negative assessments, and the only reverse transitions complete a run of k_2+1 positive assessments. Balancing these two fluxes gives $v^{k_1+1}\pi_{\mathrm{G}}=u^{k_2+1}\pi_{\mathrm{B}}$, and normalization yields

$$\pi_{\mathrm{G}}=\frac{u^{k_2+1}}{Z(u)},\quad \pi_{\mathrm{W}_m}=\frac{v^m u^{k_2+1}}{Z(u)},\quad \pi_{\mathrm{B}}=\frac{v^{k_1+1}}{Z(u)},\quad \pi_{\mathrm{S}_n}=\frac{u^n v^{k_1+1}}{Z(u)},\qquad Z(u)=v^{k_1}+u^{k_2}-v^{k_1}u^{k_2} \tag{M5}$$

(Supplementary Note 4). Averaging the readout probabilities over this distribution defines the reputation map, the stationary probability that the target is regarded as good when it is assessed positively with probability u:

$$\mathscr{R}(u)\equiv\mathscr{R}(u;\alpha,\beta,k_1,k_2)=\sum_{\ell\in\mathscr{L}}\pi_\ell(u)\,\sigma(\ell)=\frac{u^{k_2+1}\sum_{m=0}^{k_1}(\alpha v)^m+v^{k_1+1}\sum_{n=1}^{k_2}u^n\left(1-\beta^n\right)}{v^{k_1}+u^{k_2}-v^{k_1}u^{k_2}}. \tag{M6}$$

Empty sums are zero, and $\mathscr{R}(0)=0$ and $\mathscr{R}(1)=1$ by continuity. Without inertia, $\mathscr{R}(u)=u$. For one-step inertia, $\mathscr{R}(u)=\{u^2[1+\alpha(1-u)]+(1-\beta)u(1-u)^2\}/(1-u+u^2)$, and with full readout, $\mathscr{R}(u)=u^{k_2}(1-v^{k_1+1})/Z(u)$. The map increases with α and decreases with β. Two properties are used repeatedly below: for $k_1\geq 1$, $\mathscr{R}'(1)=1-\alpha$, where the prime denotes $\mathrm{d}/\mathrm{d}u$; and for $k_2=0$, $\mathscr{R}$ is concave on $[0,1]$ (Supplementary Note 4).

Self-consistent private reputations

Let r_I be the probability that a randomly chosen observer regards a randomly chosen type-I individual as good, and $\bar{r}=\sum_I f_I r_I$. We close the private-assessment problem with two mean-field assumptions, as in previous analyses of private assessment [1, 3, 5, 6, 7]. First, the donor's and the observer's images of a recipient are independent given the recipient's strategy. Both then regard a random recipient as good with probability $h_{\mathrm{GG}}=\sum_J f_J r_J^2$, both regard it as bad with probability $h_{\mathrm{BB}}=\sum_J f_J(1-r_J)^2$, and each of the two ordered disagreements occurs with probability $h_{\mathrm{GB}}=\sum_J f_J r_J(1-r_J)$. A type-$I$ donor is therefore assessed positively with probability

$$\begin{aligned} u_{\mathrm{ALLC}}&=\Phi_{\mathrm{C}}(\bar{r}),\qquad u_{\mathrm{ALLD}}=\Phi_{\mathrm{D}}(\bar{r}),\\ u_{\mathrm{DISC}}&=h_{\mathrm{GG}}P^{+}_{\mathrm{C}|\mathrm{G}}+h_{\mathrm{GB}}\left(P^{+}_{\mathrm{C}|\mathrm{B}}+P^{+}_{\mathrm{D}|\mathrm{G}}\right)+h_{\mathrm{BB}}P^{+}_{\mathrm{D}|\mathrm{B}}. \end{aligned} \tag{M7}$$

Second, the assessments entering a record of a type-I individual are treated as independent with this common probability u_I, so that each record has the stationary response of Eq. (M6). Stationary reputations then satisfy the self-consistency equations

$$r_I^{*}=\mathscr{R}\big(u_I(\boldsymbol{r}^{*};\boldsymbol{f})\big),\qquad I\in\{\mathrm{ALLC},\mathrm{ALLD},\mathrm{DISC}\}. \tag{M8}$$

Equation (M8) can have several solutions. Consistent with records that start in good standing, we select at every composition the solution reached by the auxiliary relaxation

$$\frac{\mathrm{d}r_I}{\mathrm{d}\tau}=\mathscr{R}(u_I)-r_I,\qquad r_I(0)=1. \tag{M9}$$

The variable τ is a computational time, unrelated to evolutionary time t. The fixed points of Eq. (M9) coincide with the stationary states of the mean-field dynamics of the full distribution of latent states, $\mathrm{d}\boldsymbol{\pi}_I/\mathrm{d}t'=\boldsymbol{\pi}_I[T(u_I)-\mathbf{I}]$ with $r_I=\boldsymbol{\pi}_I\boldsymbol{\sigma}$

(Supplementary Note 5). Stability statements refer to Eq. (M9); where checked, they agreed with the linearization of the full latent-state system.

In a homogeneous DISC population with reputation r, a resident donor regards its recipient as good with probability r, independently of the observer, and Eq. (M7) reduces to

$$u_{\mathrm{R}}(r) = r\,\Phi_{\mathrm{C}}(r) + (1-r)\,\Phi_{\mathrm{D}}(r). \tag{M10}$$

Resident reputations are the roots of $F(r) \equiv \mathscr{R}[u_{\mathrm{R}}(r)] - r$, and their stability in Eq. (M9) is given by the sign of $F'(r)$. A fold of a resident branch satisfies $F(r) = F'(r) = 0$, that is $\mathscr{R}'[u_{\mathrm{R}}(r)]\,u'_{\mathrm{R}}(r) = 1$. We denote by r^*_{R} the highest strictly stable root, which coincides with the root reached by relaxation from $r = 1$ unless the highest root is a fold, and write $u^*_{\mathrm{R}} = u_{\mathrm{R}}(r^*_{\mathrm{R}})$, $\Phi^*_A = \Phi_A(r^*_{\mathrm{R}})$ and $\Delta\Phi^* = \Delta\Phi(r^*_{\mathrm{R}})$. Resident reputations do not depend on b/c.

Strategy evolution and cooperation

Reputations are assumed to equilibrate quickly relative to strategy change. At each strategy composition, payoffs are evaluated with the selected stationary reputations. Per unit time, in which each individual acts on average once as donor and once as recipient, the expected payoffs are

$$\begin{aligned}
\Pi_{\mathrm{ALLC}} &= (1-e_1)\big[b\,(f_{\mathrm{ALLC}} + f_{\mathrm{DISC}} r^*_{\mathrm{ALLC}}) - c\big],\\
\Pi_{\mathrm{ALLD}} &= (1-e_1)\,b\,(f_{\mathrm{ALLC}} + f_{\mathrm{DISC}} r^*_{\mathrm{ALLD}}),\\
\Pi_{\mathrm{DISC}} &= (1-e_1)\big[b\,(f_{\mathrm{ALLC}} + f_{\mathrm{DISC}} r^*_{\mathrm{DISC}}) - c\,\bar{r}^*\big],
\end{aligned} \tag{M11}$$

and strategy frequencies follow the replicator dynamics [8]

$$\frac{\mathrm{d}f_I}{\mathrm{d}t} = f_I\big(\Pi_I - \bar{\Pi}\big), \qquad \bar{\Pi} = \sum_J f_J \Pi_J. \tag{M12}$$

The intended and realized cooperation rates are $\rho_{\mathrm{int}} = f_{\mathrm{ALLC}} + f_{\mathrm{DISC}}\bar{r}^*$ and $\rho = (1-e_1)\rho_{\mathrm{int}}$, and $\bar{\Pi} = (b-c)\rho$. In a homogeneous DISC population, $\rho_{\mathrm{int}} = r^*_{\mathrm{R}}$; realized cooperation cannot exceed $1 - e_1 = 0.98$.

To compare parameter settings globally, we integrate Eq. (M12) from $n_{\mathrm{init}} = 512$ initial compositions sampled uniformly by area from the simplex, using the same compositions for all settings that are compared. In every scan, each trajectory ended either in the zero-cooperation set (the ALLD vertex together with the low-reputation segment of the ALLD–DISC edge, along which the dynamics drift slowly towards ALLD; final $\rho \le 7\times10^{-4}$) or at an attractor with positive cooperation (final $\rho \ge 0.34$, and 0.974–0.980 in nearly all conditions). We therefore define the cooperative basin as the set of initial compositions whose trajectories reach an attractor with positive cooperation, and classify a trajectory as cooperative if its final $\rho \ge 0.01$; any threshold inside the observed gap gives the same classification, and the gap was checked for every condition. We report the mean long-run cooperation $\bar{\rho}$ (the average of the final ρ over initial compositions), the cooperative-basin size ω_{C} (the fraction of cooperative trajectories) and the cooperation level $\bar{\rho}_{\mathrm{C}}$ within the basin, with $\bar{\rho} \approx \omega_{\mathrm{C}}\bar{\rho}_{\mathrm{C}}$. For the total-depth comparison, the envelope at total depth K is the largest $\bar{\rho}$ among the $K+1$ allocations with $k_1 + k_2 = K$.

Invasion of a discriminator population

We consider a homogeneous DISC population at r^*_{R} and a rare mutant X that uses the residents' norm and record mechanism. Because the mutant is rare, it interacts with residents only and leaves their reputations and assessments unchanged. Suppose that X intends to help a resident recipient with probability χ_{X}, independently of the observer's image of that recipient, and acts independently across interactions. It is then assessed positively with probability $u_{\mathrm{X}} = \chi_{\mathrm{X}}\Phi^*_{\mathrm{C}} + (1-\chi_{\mathrm{X}})\Phi^*_{\mathrm{D}}$, its stationary reputation is $r^*_{\mathrm{X}} = \mathscr{R}(u_{\mathrm{X}})$, and its normalized payoff advantage over residents is (Supplementary Note 7)

$$\Lambda_{\mathrm{X}} \equiv \frac{\Pi_{\mathrm{X}} - \Pi_{\mathrm{R}}}{c(1-e_1)} = \big(r^*_{\mathrm{R}} - \chi_{\mathrm{X}}\big) - \frac{b}{c}\big(r^*_{\mathrm{R}} - r^*_{\mathrm{X}}\big), \tag{M13}$$

the cooperation cost saved minus the benefit lost through reputation. The resident itself corresponds to $\chi = r^*_{\mathrm{R}}$ and $u = u^*_{\mathrm{R}} = r^*_{\mathrm{R}}\Phi^*_{\mathrm{C}} + (1-r^*_{\mathrm{R}})\Phi^*_{\mathrm{D}}$.

Unconditional mutants. ALLD ($\chi = 0$) has reputation $\mathscr{R}(\Phi^*_{\mathrm{D}})$ and ALLC ($\chi = 1$) has reputation $\mathscr{R}(\Phi^*_{\mathrm{C}})$, where $\Phi^*_{\mathrm{D}} = r^*_{\mathrm{R}}P^+_{\mathrm{D|G}} + (1-r^*_{\mathrm{R}})P^+_{\mathrm{D|B}}$ and Φ^*_{C} analogously. When ALLD is regarded worse and ALLC better than the residents, Eq. (M13)

implies that the discriminator population resists both mutants if and only if

$$R_{\mathrm{L}} \equiv \frac{r_{\mathrm{R}}^*}{r_{\mathrm{R}}^* - \mathscr{R}\left[r_{\mathrm{R}}^* P_{\mathrm{D|G}}^+ + (1-r_{\mathrm{R}}^*) P_{\mathrm{D|B}}^+\right]} < \frac{b}{c} < \frac{1-r_{\mathrm{R}}^*}{\mathscr{R}\left[r_{\mathrm{R}}^* P_{\mathrm{C|G}}^+ + (1-r_{\mathrm{R}}^*) P_{\mathrm{C|B}}^+\right] - r_{\mathrm{R}}^*} \equiv R_{\mathrm{U}}. \tag{M14}$$

If ALLD is not regarded worse than the residents, no b/c excludes it; if ALLC is not regarded better, it cannot invade. Because $u_{\mathrm{R}}^* = r_{\mathrm{R}}^* \Phi_{\mathrm{C}}^* + (1-r_{\mathrm{R}}^*)\Phi_{\mathrm{D}}^*$, the interval in Eq. (M14) is non-empty if and only if $r_{\mathrm{R}}^* \mathscr{R}(\Phi_{\mathrm{C}}^*) + (1-r_{\mathrm{R}}^*)\mathscr{R}(\Phi_{\mathrm{D}}^*) < \mathscr{R}(u_{\mathrm{R}}^*)$, that is, if $\mathscr{R}$ at the resident's assessment probability lies above the chord joining its values at Φ_{D}^* and Φ_{C}^*. Without inertia, $\mathscr{R}$ is linear and $R_{\mathrm{L}} = R_{\mathrm{U}} = 1/\Delta\Phi^*$, so no benefit-to-cost ratio protects a discriminator population (Supplementary Note 7). Numerically, we classify local outcomes from the signs of the unnormalized payoff differences $\lambda_{\mathrm{ALLD}} = \Pi_{\mathrm{ALLD}} - \Pi_{\mathrm{R}}$ and $\lambda_{\mathrm{ALLC}} = \Pi_{\mathrm{ALLC}} - \Pi_{\mathrm{R}}$ (with $c = 1$), which also covers cases without finite thresholds: ALLD is not excluded if $\lambda_{\mathrm{ALLD}} \geq -10^{-10}$; otherwise ALLC invades if $\lambda_{\mathrm{ALLC}} > 10^{-10}$; otherwise the residents resist both.

Opportunistic discriminators. An opportunistic discriminator (O-DISC) behaves like DISC, except that each intended cooperation towards a recipient it regards as good is replaced, independently and before implementation error, by intended defection with probability $\delta \in [0,1]$. Thus $\chi = (1-\delta) r_{\mathrm{R}}^*$, and $\delta = 0$ and $\delta = 1$ recover DISC and ALLD. The mutant's positive-assessment probability and reputation are

$$u_{\mathrm{X}}(\delta) = u_{\mathrm{R}}^* - \delta\, r_{\mathrm{R}}^* \Delta\Phi^*, \qquad r_{\mathrm{X}}^*(\delta) = \mathscr{R}\left[u_{\mathrm{X}}(\delta)\right], \tag{M15}$$

and Eq. (M13) gives the normalized invasion payoff and its slope at the resident,

$$\Lambda(\delta) = \delta\, r_{\mathrm{R}}^* - \frac{b}{c}\left[r_{\mathrm{R}}^* - r_{\mathrm{X}}^*(\delta)\right], \qquad \Lambda'(0) = r_{\mathrm{R}}^*\left[1 - \frac{b}{c}\Delta\Phi^* \mathscr{R}'(u_{\mathrm{R}}^*)\right]. \tag{M16}$$

A positive $\Lambda(\delta)$ means that the mutant can invade; $\Lambda(0) = 0$, and $\Lambda(1)$ is the normalized ALLD advantage. For $r_{\mathrm{R}}^* > 0$, $\Lambda(\delta) = \delta r_{\mathrm{R}}^* [1 - (b/c) H(\delta)]$, where $H(\delta) = [r_{\mathrm{R}}^* - r_{\mathrm{X}}^*(\delta)]/(\delta r_{\mathrm{R}}^*)$ is the reputation lost per unit of withheld cooperation, extended continuously by $H(0) = \Delta\Phi^* \mathscr{R}'(u_{\mathrm{R}}^*)$. The residents therefore exclude every O-DISC mutant if and only if

$$\min_{0 \leq \delta \leq 1} \frac{r_{\mathrm{R}}^* - r_{\mathrm{X}}^*(\delta)}{\delta\, r_{\mathrm{R}}^*} \geq \frac{c}{b}, \tag{M17}$$

with equality permitting neutral mutants. Because $H(\delta)$ equals $\Delta\Phi^*$ times the slope of the chord of $\mathscr{R}$ between $u_{\mathrm{X}}(\delta)$ and u_{R}^*, for $\Delta\Phi^* > 0$ the minimum is attained at $\delta \to 0$ whenever $\mathscr{R}$ is concave on that interval (always for $k_2 = 0$), and Eq. (M17) then reduces to $\Lambda'(0) \leq 0$; in general, the whole interval must be checked. Near an all-good resident with $k_1 \geq 1$, $\mathscr{R}'(u_{\mathrm{R}}^*) \approx 1 - \alpha$, so full tolerance readout makes weak opportunism cheap. In the error-free limit with $\beta = 1$ and $k_1 \geq 1$, an all-good resident excludes every O-DISC mutant if and only if $\alpha \leq 1 - c/b$, for every norm and every k_2 (Supplementary Note 8); at $b/c = 5$ this bound is the readout $\alpha = 0.8$ used in Fig. 5 and Extended Data Fig. 9.

For fixed norm, depths, β and errors, a readout α is secure if a strictly stable high-reputation resident exists and satisfies Eq. (M17). The lower boundary α_{L} of a secure interval, when it is not $\alpha = 0$, is a fold of the high-reputation branch, $F(r,\alpha) = \partial_r F(r,\alpha) = 0$ with $F(r,\alpha) = \mathscr{R}[u_{\mathrm{R}}(r); \alpha, \beta, k_1, k_2] - r$; the fold itself is excluded and does not depend on b/c. The upper boundary α_{U}, when it is not $\alpha = 1$, is where Eq. (M17) holds with equality. In every case we computed it was determined by $\delta \to 0$, that is by $(b/c)\Delta\Phi(r)\mathscr{R}'[u_{\mathrm{R}}(r); \alpha_{\mathrm{U}}] = 1$ together with $F(r, \alpha_{\mathrm{U}}) = 0$. When the fold and the local condition meet, the secure interval closes; this happens at $b/c = u_{\mathrm{R}}'(r)/\Delta\Phi(r)$ evaluated at the fold, which equals 2 for Shunning (Supplementary Note 8).

Alternative reputation filters. For Fig. 5a we replace $\mathscr{R}$ by the stationary response of two alternative mechanisms while keeping the norm, errors, resident selection and Eqs. (M15)–(M16) unchanged. Aggregation, adapted from ref. [6], assigns a good image if at least Q of M independently sampled assessments are positive, $\mathscr{R}_{\mathrm{agg}}(u) = \sum_{j=Q}^{M} \binom{M}{j} u^j (1-u)^{M-j}$, with $(M,Q) = (2,1)$, so that $\mathscr{R}_{\mathrm{agg}}(u) = 2u - u^2$. Quantitative assessment, adapted from ref. [9], keeps an integer score $z \in \{-z_{\max}, \ldots, z_{\max}\}$ that moves up or down by one after a positive or negative assessment (attempts to leave the range leave the score unchanged) and is read as good if $z \geq z_0$; with $t = u/(1-u)$, $\mathscr{R}_{\mathrm{score}}(u) = \sum_{j=z_{\max}+z_0}^{2z_{\max}} t^j \big/ \sum_{j=0}^{2z_{\max}} t^j$, and we use $z_{\max} = 5$, $z_0 = 0$. Neither mechanism was optimized or matched in memory to the inertial record.

Finite-population reputation simulations

Individual-based simulations test reputation formation without the mean-field closure, at fixed strategy composition; they do not simulate strategy evolution. A population of N individuals contains fixed numbers n_I of each strategy. Every DISC

individual stores a latent record of every individual, including itself, and all records start in G. Records held by ALLC and ALLD individuals are not stored because they never affect actions or assessments. In each event: (i) a donor is drawn uniformly from the population and a recipient uniformly from the remaining $N-1$ individuals, and an observer is drawn independently and uniformly from the DISC individuals (the observer may coincide with donor or recipient); (ii) the donor forms its intended action, a DISC donor by sampling its own image of the recipient from Eq. (M4); (iii) implementation error is applied; (iv) the observer independently samples its own image of the recipient, applies the norm to the realized action and reverses the assessment with probability e_2; and (v) the observer's record of the donor is updated. A sweep consists of $n_{\mathrm{DISC}}N$ events, one update per stored record on average.

At sampled times we use $s_{ij}=\sigma(L_{ij})$, which averages exactly over readout randomness. For target j, the mean goodness and the probability that two distinct DISC observers give different images are

$$g_j=\frac{1}{n_{\mathrm{DISC}}}\sum_i s_{ij},\qquad d_j=\binom{n_{\mathrm{DISC}}}{2}^{-1}\sum_{i<i'}\left[s_{ij}(1-s_{i'j})+(1-s_{ij})s_{i'j}\right], \tag{M18}$$

where i,i' run over DISC observers. Averaging over type-I targets gives g_I and d_I, which are then averaged over sampling times. The reputational separation and mean disagreement are

$$\Delta g=\frac{g_{\mathrm{DISC}}+g_{\mathrm{ALLC}}}{2}-g_{\mathrm{ALLD}},\qquad \bar{d}=\sum_I\frac{n_I}{N}d_I. \tag{M19}$$

Unlike the mean-field theory, the simulations retain target-specific variation and correlations between the private records of different observers.

The recognition analyses use equal strategy numbers, $\alpha=\beta=1$ and $0\le k_1,k_2\le 6$. Each run lasts 1,000 sweeps and is sampled every five sweeps from sweep 800 to sweep 1,000 (41 samples). The depth maps in Fig. 2e–h use one run per condition with $N=150$. The replicated analysis (Extended Data Fig. 1) uses 30 independent runs per condition with $N=150$ and reports means across runs, with pointwise 95% percentile intervals from 2,000 bootstrap resamples of the run means. A condition is eligible for selection if, for both Δg and $\bar{d}$, the bootstrap interval of the difference between the second- and first-half means of the sampling window lies within $(-0.01,0.01)$. Two slowly relaxing SJ conditions required longer runs: $(k_1,k_2)=(4,5)$ with 3,000 sweeps (sampled from sweep 2,400) and $(3,5)$ with 50,000 sweeps (sampled from sweep 40,000); all 196 conditions then passed. Among eligible conditions with $\Delta g\ge 0.8$ and $\bar{d}\le 0.1$, we select the smallest K, breaking ties by larger Δg, then smaller $\bar{d}$, k_1 and k_2. This gives the four architectures listed above. The stars in Fig. 2e–h apply the same cutoffs and tie-breaking rule to the single-run maps, without the drift criterion, and identify the same four architectures. Selection uses reputation statistics only. The robustness analysis varies the separation cutoff from 0.70 to 0.90 and the disagreement cutoff from 0.050 to 0.150 in steps of 0.05 and 0.025. The illustrative image matrices in Fig. 2a–d are final states of single runs with $N=90$ after 1,000 sweeps.

Numerical procedures and figure-specific settings

All panels use $b/c=5$ and $e_1=e_2=0.02$ unless an axis varies them; Supplementary Note 12 (Supplementary Table S11) lists every setting, grid and tolerance. $\mathscr{R}$ and $\mathscr{R}'$ are evaluated in closed form. Homogeneous-DISC roots (Fig. 5; Extended Data Figs. 2a–d, 3 and 9) are located by evaluating $F(r)$ on a fine uniform grid, refining sign changes by bracketing and classifying roots by the sign of $F'(r)$. At mixed compositions, Eq. (M9) is integrated from $r_I=1$ by fourth-order Runge–Kutta steps until $\max_I|\mathscr{R}(u_I)-r_I|\le 10^{-11}$ (10^{-12} for Fig. 3). For local invasion (Extended Data Figs. 5–8), the resident relaxation is followed by step-limited Newton refinement, which keeps the branch reached from $r=1$, to residual $\le 10^{-13}$.

All sampled-cooperation results (Fig. 4; Extended Data Figs. 2e, 5, 7 and 8e–h) use one engine. Selected reputations are tabulated exactly on a triangular lattice with 120 subdivisions and interpolated linearly only inside cells in which no reputation component varies by more than 0.02; elsewhere, including every cell crossed by a fold, they are computed exactly at the requested composition. Payoffs and the replicator field are always computed from these reputations, never interpolated. Trajectories are integrated with the explicit midpoint rule ($\Delta t=0.2$) until ρ_{int} changes by at most 10^{-7} over each of three consecutive intervals of 100 time units, or for at most 4×10^4 time units. Halving the step, refining the lattice and tightening the tolerances changed $\bar{\rho}$ by at most 1.3×10^{-5} and no outcome class. Initial compositions are 512 points of a randomly shifted Halton sequence (bases 2 and 3; seed 2026) or, for Extended Data Fig. 5, eight independently scrambled 512-point Sobol sequences [10] (seeds 2026–2033), mapped area-uniformly onto the simplex by reflecting points of the unit square with $x+y>1$. In Extended Data Fig. 5, error bars are standard errors across the eight samples and therefore describe the sampling of initial compositions; the dynamics are deterministic. Extended Data Fig. 2e initializes reputations once, at $t=0$, as all-good

or all-bad and then follows them quasi-statically along each trajectory by warm-started relaxation, so that a branch is left only where it ends.

In Fig. 3 and Extended Data Fig. 8a–d, reputations are relaxed exactly at every point of an 84×74 grid clipped to the simplex. Vertex stability follows from the two rare-strategy growth rates; edge equilibria are located on 501-point edge scans, and their markers refer to stability along the edge; interior equilibria are classified by their Jacobian. Under SC without inertia, all three payoffs are equal on the line $f_{\mathrm{DISC}} = c/[b(\varepsilon - e_2)] \approx 0.2126$ (Supplementary Note 6). Extended Data Fig. 3 reports the highest stable homogeneous-DISC root and marks bistability (one-step readout plane, 161×161; depth plane, $0 \le k_1, k_2 \le 8$); for every root in the depth planes, stability in Eq. (M9) agreed with the linearization of the full latent-state dynamics. On the ALLD–DISC edge (Extended Data Fig. 4), Eq. (M8) reduces exactly to one equation in $\bar{r}$, which is solved for 1,071 values of $f_{\mathrm{DISC}} \in [0.01, 1]$. Local invasion maps use 241×241 points of $e_1, e_2 \in [0, 0.45]$ (Extended Data Fig. 6) and 801×801 points of (α, β) (Extended Data Fig. 8i–l). In Fig. 5, $\Lambda(\delta)$ is evaluated at 5,015 values of δ, concentrated near $\delta = 0$, with sign changes refined by Brent's method; δ_{c} is the upper end of the interval with $\Lambda > 0$. Residents of the alternative filters are obtained from $r = 1$ by damped iteration, $r \leftarrow 0.8r + 0.2\mathscr{R}_{\mathrm{X}}[u_{\mathrm{R}}(r)]$ with $\mathscr{R}_{\mathrm{X}} \in \{\mathscr{R}_{\mathrm{agg}}, \mathscr{R}_{\mathrm{score}}\}$. For Extended Data Fig. 9 (≥ 601 values of α, 161 values of $b/c \in [2, 8]$, $\beta = 1$), $\min_\delta H(\delta)$ is found by comparing H at $\delta = 0$, $\delta = 1$ and all real roots in $(10^{-8}, 1)$ of the numerator of the rational function $\mathrm{d}H/\mathrm{d}\delta$, and a readout is classified as secure if $1 - (b/c)\min_\delta H \le 10^{-10}$. Fold endpoints are excluded; endpoints at the domain boundary or at neutral invasion are included.

Data and code availability

The code, numerical data and figure scripts used in this study are provided with the paper (verified computational package; one notebook per figure, a C engine for the strategy dynamics, input configurations, validation records and checksums). *[Repository and DOI to be added before submission.]*

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